EV Technology Course — Module 9: Brakes, Steering and Suspension (By-Wire Systems)

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Without an engine, an EV has no engine vacuum for the brake booster and no engine-driven hydraulic pump for steering, so both became electric. Once they are electric, the next step is "by-wire": the pedal and steering wheel send signals, and actuators do the work. This enables smooth regen blending and is a foundation for automated driving, but it demands redundancy because nothing mechanical is left as a backup.

1. Learning objectives

- Explain why EVs need electric brake boosters and how brake blending works.
- Compare vacuum, electromechanical booster, integrated "one-box" and electromechanical brakes.
- Compare EPS types and explain steer-by-wire.
- Explain redundancy and fail-operational design for safety-critical by-wire systems.
- Describe EV-specific suspension and brake issues (mass, corrosion, NVH).

2. Fundamentals

System	Petrol car	EV
Brake booster	Vacuum from engine intake	Electric booster or integrated brake system
Braking energy	All wasted as heat	Mostly recovered by regen; friction tops up
Steering assist	Hydraulic pump (older) or EPS	Electric power steering (EPS); steer-by-wire emerging
Suspension	Tuned for engine mass up front	Tuned for heavy, low battery; often air or adaptive dampers in premium

By-wire means the driver's input is measured by sensors and executed by electric actuators, with no direct mechanical (or hydraulic) connection in normal operation.

3. Detailed technical explanation

Braking systems

Type	How it works	Regen blending	Status
Vacuum booster + electric vacuum pump	Keeps the old design; pump replaces engine vacuum	Limited	Low-cost and early EVs
Electromechanical booster (e.g. Bosch iBooster)	Electric motor drives the master cylinder	Good: can decouple pedal feel	Common
Integrated brake / one-box (booster + ESC in one unit)	Pedal simulator; motor-driven pressure generation	Excellent; brake-by-wire with hydraulic fallback	Increasingly common
Electromechanical brakes (EMB)	Electric motor at each caliper; no brake fluid	Excellent	Early production / pilot

Steering systems

Type	Notes
Column-assist EPS	Motor on steering column; small cars
Pinion/dual-pinion EPS	Mid-size cars
Rack-assist (belt-drive) EPS	Heavy cars and EVs; high assist forces
Steer-by-wire	No mechanical shaft: steering wheel has a feedback (feel) actuator; rack has redundant motors and ECUs

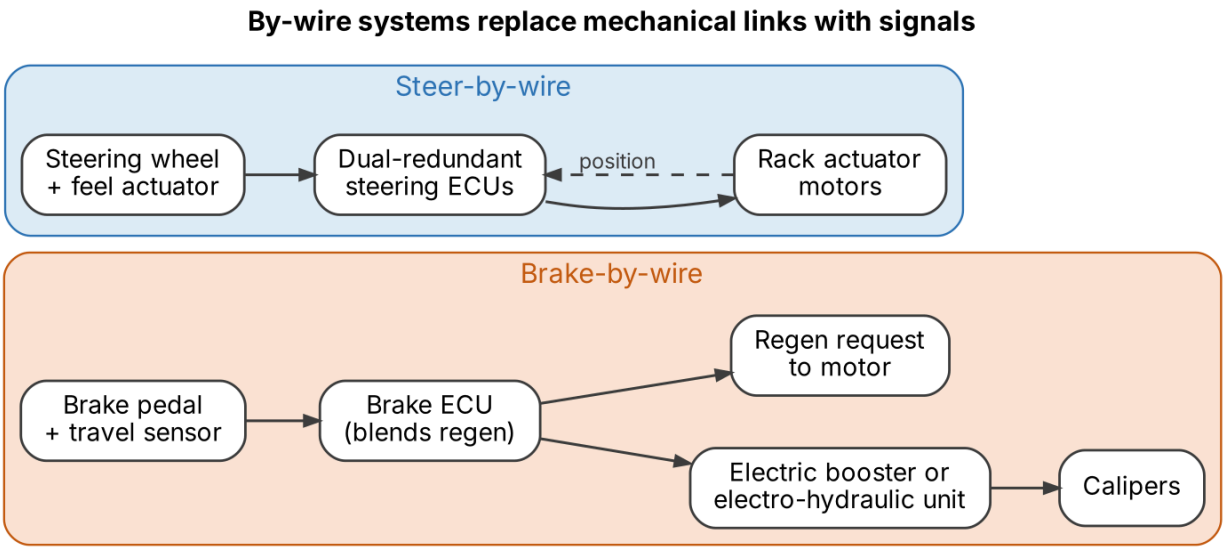
Redundancy and safety. By-wire braking and steering are rated ASIL D (the highest) under ISO 26262. Designs use duplicated sensors, power supplies, processors, communication and actuator windings, so that a single fault leaves the system able to keep working (fail-operational) or bring the car to a safe state. Regulations such as UNECE R79 (steering) and R13-H (braking) govern approval.

EV-specific issues

- **Disc corrosion:** regen does most braking, so friction brakes are used rarely and discs rust; software may periodically "clean" discs with friction braking. Some EVs use rear drum brakes for this reason.

- **Mass:** heavier EVs need larger brakes for emergency and fade cases even though everyday use is lighter.
- **NVH:** creaks and clunks are more noticeable without engine noise.
- **Suspension:** stiffer springs and re-tuned dampers for higher mass; air suspension in premium EVs to adjust ride height for aero and ground clearance.

4. Diagram: by-wire systems replace mechanical links with signals



5. Step-by-step: a brake-by-wire stop with a fault

1. Driver presses the pedal; a pedal-travel sensor and a pedal simulator give natural feel.
2. The brake ECU calculates the deceleration requested and asks the VCU for maximum regen.
3. The motor-driven pressure unit builds hydraulic pressure for the remainder.
4. Suppose the primary ECU fails mid-stop: the secondary channel (separate power and processor) takes over within milliseconds.
5. If all electronics fail, a hydraulic fallback lets the pedal push fluid directly (in one-box designs), or redundant EMB actuators still stop the car.
6. The fault is logged and the driver is warned.

6. Engineering equations

Stopping distance (constant deceleration):

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d = \frac{v^2}{2a}
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Brake torque from one caliper:

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T_b = 2 \mu p A_p r_{eff}
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Steering assist power:

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P = F_{rack} \times v_{rack}
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Reliability of two independent channels:

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R_{sys} = 1 - (1 - R)^2
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μ = pad friction coefficient, p = hydraulic pressure (Pa), A_p = piston area (m²), r_eff = effective disc radius (m).

7. Worked examples

Example 1: stopping distance. From 100 km/h at 0.9 g: $27.8^2 \div (2 \times 8.83)$ = **43.7 m** (plus reaction distance).

Example 2: caliper torque. μ = 0.4, 80 bar (8 MPa), piston area 25 cm², r_eff = 0.13 m: $T = 2 \times 0.4 \times 8 \times 10^6 \times 0.0025 \times 0.13$ = **2,080 N·m** per wheel.

Example 3: redundancy. If one channel fails with probability 10^{-5} per hour, two independent channels fail together with probability 10^{-10} per hour. The key word is *independent*: shared power supplies or software bugs break this assumption.

8. Real-world examples

- **Electromechanical boosters / one-box systems** are widely used in EVs (Bosch iBooster and IPB, Continental MK C1/C2, ZF IBC and others).
- **Steer-by-wire in production:** Tesla Cybertruck, Lexus RZ (selected markets) and some Chinese premium models; check model-year availability.
- **Rear drum brakes:** used on some EVs (e.g. VW ID.3 / ID.4 rear) because regen does most braking and drums resist corrosion.
- **Indian EVs:** electric boosters and EPS are standard; steer-by-wire is not yet offered.

9. Manufacturing applications

- Brake and steering systems arrive as tested modules from tier-1s; the plant fills and bleeds brakes, calibrates steering angle sensors, and runs ESC/ABS function tests on rollers.
- Software flashing and calibration per variant; end-of-line checks of brake blending on a dyno.
- Costing: one-box brakes cost more than vacuum systems but remove vacuum pump and allow regen recovery (range value); steer-by-wire adds actuators and redundancy but removes the steering column and enables flexible cabin layouts.

10. Common mistakes and misconceptions

- **"By-wire means less safe."** Properly designed redundancy can meet or exceed mechanical reliability.
- **"EV brakes wear out faster because of weight."** Usually the opposite: regen does most braking, so pads last much longer; corrosion is the bigger issue.
- **"Regen replaces ABS."** ABS and ESC still modulate friction brakes in emergencies.
- **"Steer-by-wire is just EPS."** EPS keeps a mechanical shaft; steer-by-wire removes it.

11. Practical case study: corroded discs at service

Problem. EV owners fail roadworthiness checks because rear discs are badly corroded after two years.

Root cause. High regen share meant rear friction brakes almost never engaged; coastal humidity accelerated rust.

Fix. Software "brake cleaning" cycles using light friction braking periodically, coated discs, or drum brakes on the rear axle; service guidance.

Lesson. Regen changes not only energy use but the failure modes of other systems.

12. Revision notes

- No engine vacuum → electric booster or integrated brake.
- Brake-by-wire enables seamless regen blending.
- EPS standard; steer-by-wire removes the shaft and needs redundancy.
- ASIL D, fail-operational designs with independent channels.
- EV brake issue: corrosion, not wear.

13. Glossary

Term	Meaning
iBooster / electromechanical booster	Motor-driven brake booster
One-box / IPB	Integrated booster + ESC unit
EMB	Electromechanical brake (no hydraulics)
EPS	Electric power steering
Pedal simulator	Gives brake-pedal feel when pedal is decoupled
Fail-operational	Keeps working after a single fault
ASIL D	Highest automotive safety integrity level

14. Knowledge check

Conceptual

1. Why can't EVs use a conventional vacuum booster alone?
2. Why is brake-by-wire useful for regen?
3. What must be redundant in a steer-by-wire system?
4. Why do EV brake discs corrode?
5. What does steer-by-wire enable for future vehicles?

Numerical

1. Stopping distance from 80 km/h at 0.8 g?
2. EPS rack force 8,000 N at rack speed 0.05 m/s: assist power?
3. Two independent channels, each 99.9% reliable over a mission: system reliability?

Application

1. Compare the cost and benefits of an iBooster vs a one-box system for an Indian EV.
2. Outline validation tests for a steer-by-wire system.

15. Answer key

Conceptual

1. There is no engine intake vacuum; an electric vacuum pump works but adds noise, power draw and limits blending.

2. The pedal is decoupled from hydraulic pressure, so the system can swap regen for friction without changing pedal feel.
3. Sensors, ECUs, power supplies, communication, and actuator motors/windings, with independent failure paths.
4. Friction brakes are rarely used, so rust is not wiped off; humidity and salt make it worse.
5. Flexible cabin layouts (retractable wheel), variable steering ratios, and direct control for automated driving.

Numerical

1. $22.2^2 \div (2 \times 7.85) =$ **31.5 m**.
2. $8{,}000 \times 0.05 =$ **400 W**.
3. $1 - 0.001^2 =$ **99.9999%**.

Application

1. iBooster: proven, lower cost, works with existing ESC; good blending. One-box: fewer components, better packaging and blending, better for future automation, higher unit cost and supplier lock-in. Decide on volume, platform plans and supplier support.
2. Fault-injection (HIL) for every single and relevant double fault, EMC and power-supply dropouts, durability and environmental tests, vehicle tests (lane change, emergency manoeuvres, low-mu), cybersecurity tests, homologation to UNECE R79.

16. Further learning

- Bosch Automotive Handbook: braking and steering.
- ISO 26262 Part 5 (hardware) and fail-operational design papers.
- UNECE R13-H and R79.
- SAE papers on electromechanical brakes and steer-by-wire.

Next: Module 10, E/E architecture, software and connectivity.